AlgebraDx

Open Diagnostic Item Bank for Algebra Readiness

Teacher guide · Grades 9–12 · Version 1.0.0

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40 original multiple-choice items in 8 strands. Every wrong answer is written to reveal one of 64 documented misconceptions, and every misconception comes with a teaching move. No student data is collected. Schools calibrate the items on their own students with the included kit.

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How to use this bank

1. Give a form (20 minutes). Forms A and B are parallel 20-item checks covering all eight strands. Use one as a start-of-unit diagnostic and the other after re-teaching. Tell students the check is not graded and that a careful guess is useful.

2. Read the wrong answers, not just the score. Each wrong option points to a specific misconception code. A student who chooses two or more options with the same code is showing a pattern worth acting on. One isolated choice may be a slip.

3. Confirm with the probe. Before re-teaching, ask the one-line probe question listed for that code in Appendix A. A 30-second conversation separates a misconception from a careless error.

4. Teach the move, then re-check. Each item and each code carries a teaching move drawn from the research. Use the parallel form, or any other item with the same code, to check the change.

5. Calibrate locally (optional). The difficulty levels in this guide are the author's hypotheses. To get real difficulty and discrimination values for your students, enter responses in calibration/response_template.csv and run calibration/calibrate.py. Data stays in your school.

The eight strands

Strand	Why it matters for readiness	Items
EQ Equality and equivalence	Students who read "=" as "the answer comes next" cannot reason about equivalent equations, which is the core move of all equation solving.	EQ01, EQ02, EQ03, EQ04, EQ05
VAR Variables and expressions	Algebra requires letters to stand for numbers that vary. Students who treat letters as labels or single unknowns stall when expressions must be compared or generalized.	VAR01, VAR02, VAR03, VAR04, VAR05
DIST Distributive property and factoring	Expanding and factoring are the transformations of algebra. Errors here (including over-applying "linearity") recur in every later course.	DIST01, DIST02, DIST03, DIST04, DIST05
LIN Linear equations and inequalities	Solving linear equations and inequalities is the gateway skill of Algebra I, and the place where move-it rules replace reasons.	LIN01, LIN02, LIN03, LIN04, LIN05
NUM Integers, exponents and rational expressions	Many "algebra" errors are number-sense errors in disguise, especially with negatives, exponents and fractions.	NUM01, NUM02, NUM03, NUM04, NUM05
LINF Linear functions and rate of change	Slope as a rate and the meaning of parameters in context connect symbols, tables and graphs, the bridge into functions.	LINF01, LINF02, LINF03, LINF04, LINF05
FUNC Functions and notation	Function notation and the definition of a function are new in high school, and misreadings (f(x) as multiplication) block Algebra II.	FUNC01, FUNC02, FUNC03, FUNC04, FUNC05
MOD Translating words into symbols	Modeling requires moving from relationships described in words to symbols. Keyword strategies and word-order translation fail here.	MOD01, MOD02, MOD03, MOD04, MOD05

EQ · Equality and equivalence

EQ01 · Equality and equivalence

Difficulty 1 (hypothesized) · Form A #1

Diagnoses: Recognize equations that are equivalent to a given linear equation · **CCSS:** A-REI.A.1, 8.EE.C.7 · **NAEP focus:** equations and inequalities - equivalent forms

Without solving, which equation has the same solution as $3x + 7 = 22$?

Choice	What it suggests	Misconception
A $3x = 15$	**Correct.**	
B $3x = 29$	Moved +7 to the right side without changing its sign ($22 + 7$).	**EQ-TRANSPOSE** "Move it to the other side" without the inverse operation
C $x + 7 = 19$	Read $3x$ as $3 + x$ and 'removed' the 3 by subtracting it from 22.	**VAR-COEF-ADD** Coefficient read as addition ($3x$ as $3 + x$)
D $3x + 7 = 15$	Subtracted 7 from the right side only, leaving +7 on the left.	**EQ-ONESIDE** Operation applied to only part of the equation

Teaching move. Have students substitute $x = 5$ into each choice; an equation is equivalent only if the same value still makes it true. Then name the both-sides operation that produced the correct one.

EQ02 · Equality and equivalence

Difficulty 1 (hypothesized) · Form A #2

Diagnoses: Use the equal sign relationally (compare structure without computing) · **CCSS:** A-SSE.A.2, 6.EE.A.4 · **NAEP focus:** variables, expressions and operations - equivalence and structure

Without adding, decide whether this statement is true or false: $47 + 38 = 48 + 37$.

Choice	What it suggests	Misconception
A False, because the numbers on the two sides are different.	Judges equality by whether the symbols match rather than whether the values match.	**VAR-SURFACE** Judging by surface features of symbols
B False, because $47 + 38 = 85$, and 85 is not 48.	Reads '=' as 'the answer is next' and treats 48 as the result of $47 + 38$.	**EQ-OP** Equal sign as "the answer comes next"
C True, because 48 is one more than 47 and 37 is one less than 38.	**Correct.**	
D You cannot tell without computing both sides.	Can verify only by computing; does not yet use structure (compensation) to reason about equality.	**EQ-OP** Equal sign as "the answer comes next"

Teaching move. Run a short true/false number-sentence routine (5 minutes, daily for a week) in which every statement must be justified without computing.

EQ03 · Equality and equivalence

Difficulty 2 (hypothesized) · Form A #3

Diagnoses: Choose a valid first step that preserves equality · **CCSS:** A-REI.A.1, A-REI.B.3 · **NAEP focus:** equations and inequalities - solving linear equations

Which of these is a correct first step for solving $4x - 8 = 12$?

Choice	What it suggests	Misconception
A $x - 8 = 3$	Divided 4x and 12 by 4 but not the 8; division is applied to the most visible terms, not to whole sides.	**EQ-ONESIDE** Operation applied to only part of the equation
B $x - 2 = 3$	**Correct.**	
C $4x = 4$	Moved −8 across as −8 (12 − 8) instead of adding 8 to both sides.	**EQ-TRANSPOSE** "Move it to the other side" without the inverse operation
D $x - 8 = 12$	Divided only the x-term by 4, so 'getting x alone' was done to one term.	**EQ-ONESIDE** Operation applied to only part of the equation

Teaching move. Require a substitution check after each step: x = 5 solves the original, so it must solve every correct next line.

EQ04 · Equality and equivalence

Difficulty 2 (hypothesized) · Form B #1

Diagnoses: Collect variable terms from both sides of an equation · **CCSS:** A-REI.B.3, 8.EE.C.7b · **NAEP focus:** equations and inequalities - solving linear equations

Which equation is a correct next step for solving $5x + 3 = 2x + 15$?

Choice	What it suggests	Misconception
A $7x + 3 = 15$	Moved 2x to the left keeping its sign, so the x-terms were added instead of subtracted.	**EQ-TRANSPOSE** "Move it to the other side" without the inverse operation
B $5x + 3 = 17x$	Combined 2x + 15 into 17x, treating unlike terms as combinable.	**VAR-CONJOIN** Conjoining unlike terms
C $5x = 2x + 18$	Moved +3 to the right keeping its sign (15 + 3).	**EQ-TRANSPOSE** "Move it to the other side" without the inverse operation
D $3x + 3 = 15$	**Correct.**	

Teaching move. Write the both-sides operation above each step (−2x on each side) and contrast with the 'move it' versions in a correct/incorrect worked-example pair.

EQ05 · Equality and equivalence

Difficulty 3 (hypothesized) · Form B #2

Diagnoses: Interpret an identity (infinitely many solutions) · **CCSS:** A-REI.A.1, 8.EE.C.7a · **NAEP focus:** equations and inequalities - number of solutions

For which values of x is $2(x + 3) = 2x + 6$ true?

Choice	What it suggests	Misconception
A Only x = 0	Tested one convenient value and stopped, expecting exactly one solution.	**EQ-UNIQUE** Every equation has exactly one numerical solution
B No values, because the x-terms cancel	Reached 6 = 6 (or 0 = 0) and read the disappearance of x as 'no solution'.	**EQ-UNIQUE** Every equation has exactly one numerical solution
C Every real number	**Correct.**	
D Only x = −3	Looked for a single 'special' value by setting the parentheses to zero.	**EQ-UNIQUE** Every equation has exactly one numerical solution

Teaching move. Substitute several values (0, 1, 10, −3) and record the results in a table; then sort a set of equations into one / none / infinitely many solutions.

VAR · Variables and expressions

VAR01 · Variables and expressions

Difficulty 1 (hypothesized) · Form A #4

Diagnoses: Evaluate an expression with a coefficient by substitution · **CCSS:** 6.EE.A.2c, A-SSE.A.1 · **NAEP focus:** variables, expressions and operations - evaluating expressions

If $x = 6$, what is the value of $4x - x$?

Choice	What it suggests	Misconception
A 4	Treated '- x' as deleting the letter, leaving 4; x is not acting as a quantity.	**VAR-NOT-NUMBER** Letter treated as not standing for a number
B 18	**Correct.**	
C 40	Read 4x with x = 6 as the two-digit number 46, then subtracted 6.	**VAR-CONCAT** Concatenation (3x with x = 6 read as 36)
D 3	Simplified to 3x and then dropped the x (or treated it as 1).	**VAR-NOT-NUMBER** Letter treated as not standing for a number

Teaching move. Write the multiplication sign during substitution ($4 \cdot 6 - 6$), then fade it; read $4x - x$ aloud as 'four of x, take away one of x'.

VAR02 · Variables and expressions

Difficulty 1 (hypothesized) · Form A #5

Diagnoses: Combine like terms · **CCSS:** 7.EE.A.1, A-SSE.A.2 · **NAEP focus:** variables, expressions and operations - equivalent expressions

Which expression is equivalent to $3x + 2 + 5x$?

Choice	What it suggests	Misconception
A $8x + 2$	**Correct.**	
B $10x$	Added every number and attached one x; expects a single-term answer.	**VAR-CONJOIN** Conjoining unlike terms
C $15x + 2$	Multiplied the coefficients 3 and 5 when combining like terms.	**LIKE-TERMS-OP** Adding like terms treated as multiplication
D $8x^2 + 2$	Treated $x + x$ as x^2, mixing the rules for adding and multiplying terms.	**LIKE-TERMS-OP** Adding like terms treated as multiplication

Teaching move. Evaluate the original and each choice at $x = 10$: equivalent expressions give equal values for every x.

VAR03 · Variables and expressions

Difficulty 2 (hypothesized) · Form B #5

Diagnoses: Understand that different letters may take equal values · **CCSS:** 6.EE.A.2, A-SSE.A.1 · **NAEP focus:** variables, expressions and operations - meaning of variables

Is a + b + c = a + p + c always true, never true, or sometimes true?

Choice	What it suggests	Misconception
A Always true	Matched the a's and c's on both sides and stopped looking.	**VAR-SURFACE** Judging by surface features of symbols
B Never true, because b and p are different letters	Believes different letters must stand for different numbers.	**VAR-DIFFERENT** Different letters must have different values
C Sometimes true, when b = p	**Correct.**	
D Never true, because there are no numbers to check	Does not see letters as standing for numbers that can be tested.	**VAR-NOT-NUMBER** Letter treated as not standing for a number

Teaching move. Ask for values that make it true and values that make it false; establish the 'always / sometimes / never, with an example' routine.

VAR04 · Variables and expressions

Difficulty 2 (hypothesized) · Form B #3

Diagnoses: Represent a general property (evenness) with an expression · **CCSS:** A-SSE.A.1, A-CED.A.1 · **NAEP focus:** algebraic representations - expressions for general quantities

n can be any whole number. Which expression is ALWAYS an even number?

Choice	What it suggests	Misconception
A n + 2	Checked one even value of n (such as 2) and generalized.	**VAR-SPECIFIC** Generalizing from a single example
B e	Used the letter as an abbreviation for the property 'even'.	**VAR-LABEL** Letter as object or label ("e for even", "a for apples")
C 2n + 1	Saw the 2 and concluded 'even' without reasoning about the + 1.	**VAR-SURFACE** Judging by surface features of symbols
D 2n	**Correct.**	

Teaching move. Counterexample hunt: for each choice, find a whole number n that makes it odd. Then ask why no such n exists for 2n.

VAR05 · Variables and expressions

Difficulty 3 (hypothesized) · Form B #4

Diagnoses: Compare expressions over all values of the variable · **CCSS:** A-SSE.A.1, A-REI.D.11 · **NAEP focus:** variables, expressions and operations - comparing expressions

Which is larger, 2n or n + 2?

Choice	What it suggests	Misconception
A 2n, because multiplying makes a number bigger	Applies 'multiplication makes bigger' without testing small or negative values.	**VAR-SURFACE** Judging by surface features of symbols
B They are equal	Tested n = 2, where they are equal, and generalized.	**VAR-SPECIFIC** Generalizing from a single example
C n + 2, because adding 2 always gives more	Tested a small value such as n = 1 and generalized.	**VAR-SPECIFIC** Generalizing from a single example
D It depends on the value of n	**Correct.**	

Teaching move. Table of values for n = −2, 0, 1, 2, 3, 10; graph both expressions and locate where they cross.

DIST · Distributive property and factoring

DIST01 · Distributive property and factoring

Difficulty 1 (hypothesized) · Form A #6

Diagnoses: Apply the distributive property to a binomial · **CCSS:** 7.EE.A.1, A-SSE.A.2 · **NAEP focus:** variables, expressions and operations - equivalent expressions

Which expression is equivalent to $4(x + 3)$?

Choice	What it suggests	Misconception
A $4x + 3$	Multiplied only the first term inside the parentheses.	**DIST-FIRST** Distributes to the first term only
B $4x + 7$	Added the 4 to the 3 instead of multiplying.	**DIST-ADD** Adds the outside factor instead of multiplying
C $4x + 12$	**Correct.**	
D $12x$	Multiplied the numbers and attached the x, producing a single term.	**VAR-CONJOIN** Conjoining unlike terms

Teaching move. Area model: a 4 by (x + 3) rectangle has two regions, 4x and 12. Then 'four groups of (x + 3)' written out once.

DIST02 · Distributive property and factoring

Difficulty 2 (hypothesized) · Form A #7

Diagnoses: Subtract a binomial (distribute a negative) · **CCSS:** 7.EE.A.1, A-APR.A.1 · **NAEP focus:** variables, expressions and operations - operations with expressions

Which expression is equivalent to $10 - (x - 3)$?

Choice	What it suggests	Misconception
A $13 - x$	**Correct.**	
B $7 - x$	Dropped the parentheses without changing the sign of -3.	**DIST-NEG** Negative not distributed to every term
C $13 + x$	Changed the sign of the last term only; the negative was not distributed to x.	**DIST-NEG** Negative not distributed to every term
D $x - 13$	Treated subtraction as if order did not matter, reversing the whole difference.	**SIGN-SUB** Errors with subtracting or multiplying negatives

Teaching move. Rewrite as $10 + (-1)(x - 3)$ and distribute; check with $x = 0$ (the original gives 13).

DIST03 · Distributive property and factoring

Difficulty 2 (hypothesized) · Form A #8

Diagnoses: Square a binomial · **CCSS:** A-APR.A.1, A-SSE.A.2 · **NAEP focus:** variables, expressions and operations - polynomial operations

Which expression is equivalent to $(x + 5)^2$?

Choice	What it suggests	Misconception
A $x^2 + 25$	Squared each term separately, extending distribution to exponents.	**DIST-SQUARE** $(a + b)^2 = a^2 + b^2$ (linearity overgeneralization)
B $x^2 + 10x + 25$	**Correct.**	
C $x^2 + 5x + 25$	Knows there is a middle term but recorded only one of the two cross products.	**DIST-PARTIAL** Incomplete expansion (misses a cross term)
D $2x + 10$	Read 'squared' as 'times 2'.	**EXP-MULT** Exponent treated as multiplication ($x^3 = 3x$, $2^{-3} = -6$)

Teaching move. Draw the square with side $x + 5$ and label all four regions; numeric check: $(3 + 5)^2 = 64$, but $3^2 + 5^2 = 34$.

DIST04 · Distributive property and factoring

Difficulty 2 (hypothesized) · Form B #6

Diagnoses: Simplify a rational expression by factoring before cancelling · **CCSS:** A-SSE.A.2, A-APR.D.6 · **NAEP focus:** variables, expressions and operations - equivalent expressions

Which expression is equivalent to $(3x + 6) / 3$?

Choice	What it suggests	Misconception
A $x + 6$	Cancelled the 3 in $3x$ with the denominator, leaving the 6 untouched.	**DIST-CANCEL** Cancelling across addition in a fraction
B $3x + 2$	Divided only the constant term by 3.	**DIST-FIRST** Distributes to the first term only
C $3x$	Combined $3x + 6$ into $9x$, then divided.	**VAR-CONJOIN** Conjoining unlike terms
D $x + 2$	**Correct.**	

Teaching move. Factor first, then cancel: $3(x + 2) / 3$. Check at $x = 1$: the original gives 3.

DIST05 · Distributive property and factoring Difficulty 3 (hypothesized) · Form B #7

Diagnoses: Factor a difference of squares · **CCSS:** A-SSE.A.2, A-SSE.B.3a · **NAEP focus:** variables, expressions and operations - factoring

Which expression is a factored form of $x^2 - 9$?

Choice	What it suggests	Misconception
A $(x - 3)^2$	Matched the shape 'x^2 and 9' to a perfect square without checking the middle term.	**FACT-SQUARE** Difference of squares treated as a perfect square
B $(x - 3)(x + 3)$	**Correct.**	
C $(x - 9)(x + 1)$	Found factors of -9 without matching the x-coefficient (zero).	**FACT-SUM-PRODUCT** Sum/product search misapplied
D $x(x - 9)$	Factored out x as if every term contained x.	**FACT-COMMON** Pulls out a factor that is not common to every term

Teaching move. Multiply back every factorization; contrast $x^2 - 9$ with $x^2 - 6x + 9$ side by side.

LIN · Linear equations and inequalities

LIN01 · Linear equations and inequalities

Difficulty 1 (hypothesized) · Form A #9

Diagnoses: Solve a linear equation with a negative coefficient · **CCSS:** A-REI.B.3, 8.EE.C.7b · **NAEP focus:** equations and inequalities - solving linear equations

Solve $7 - 2x = 15$.

Choice	What it suggests	Misconception
A $x = 4$	Reached $-2x = 8$ and divided by 2 instead of -2.	**SIGN-DROP** Drops or mishandles a negative sign
B $x = -11$	Moved 7 across keeping its sign ($-2x = 22$).	**EQ-TRANSPOSE** "Move it to the other side" without the inverse operation
C $x = 3$	Combined $7 - 2x$ into $5x$.	**VAR-CONJOIN** Conjoining unlike terms
D $x = -4$	**Correct.**	

Teaching move. Circle each term with its sign before solving; finish every solution with a substitution check in the original.

LIN02 · Linear equations and inequalities

Difficulty 2 (hypothesized) · Form A #10

Diagnoses: Solve a linear inequality requiring division by a negative · **CCSS:** A-REI.B.3 · **NAEP focus:** equations and inequalities - solving linear inequalities

Solve $-3x + 4 > 10$.

Choice	What it suggests	Misconception
A $x < -2$	**Correct.**	
B $x > -2$	Solved exactly as an equation; did not reverse the inequality after dividing by -3.	**INEQ-FLIP** Inequality direction not reversed after multiplying or dividing by a negative
C $x > 2$	Dropped the negative coefficient ($3x > 6$), so no reversal seemed necessary.	**SIGN-DROP** Drops or mishandles a negative sign
D $x < -14/3$	Reversed correctly but moved +4 across keeping its sign ($-3x > 14$).	**EQ-TRANSPOSE** "Move it to the other side" without the inverse operation

Teaching move. Test a value ($x = -5$) in the original and in the proposed answer; derive the reversal from the number line rather than as a rule.

LIN03 · Linear equations and inequalities

Difficulty 1 (hypothesized) · Form B #10

Diagnoses: Solve a two-step equation involving division · **CCSS:** A-REI.B.3, 7.EE.B.4a · **NAEP focus:** equations and inequalities - solving linear equations

Solve $x/3 + 2 = 5$.

Choice	What it suggests	Misconception
A $x = 1$	Undid division by 3 by dividing by 3 again.	**INV-OP** Wrong inverse operation
B $x = 13$	Multiplied $x/3$ and 5 by 3 but not the 2.	**EQ-ONESIDE** Operation applied to only part of the equation
C $x = 9$	**Correct.**	
D $x = 21$	Moved +2 across keeping its sign ($x/3 = 7$).	**EQ-TRANSPOSE** "Move it to the other side" without the inverse operation

Teaching move. Arrow diagram: x −> divide by 3 −> add 2 −> 5, then undo backwards with the inverse written on each arrow.

LIN04 · Linear equations and inequalities

Difficulty 2 (hypothesized) · Form B #8

Diagnoses: Solve an equation with parentheses and the variable on both sides · **CCSS:** A-REI.B.3, 8.EE.C.7b · **NAEP focus:** equations and inequalities - solving linear equations

Solve $3(x - 2) = x + 4$.

Choice	What it suggests	Misconception
A $x = 3$	Expanded $3(x - 2)$ as $3x - 2$.	**DIST-FIRST** Distributes to the first term only
B $x = 5$	**Correct.**	
C $x = 5/2$	Moved x to the left keeping its sign ($3x + x = 10$).	**EQ-TRANSPOSE** "Move it to the other side" without the inverse operation
D There is no solution, because x is on both sides.	Believes an unknown on both sides cannot be solved.	**EQ-BOTHSIDES** Unknown on both sides cannot be solved

Teaching move. Balance model: removing x from both sides keeps the balance level; check the solution by substituting into the original.

LIN05 · Linear equations and inequalities Difficulty 2 (hypothesized) · Form B #9

Diagnoses: Solve a literal equation for a variable · **CCSS:** A-CED.A.4 · **NAEP focus:** equations and inequalities - formulas and literal equations

The formula d = rt relates distance, rate and time. Which expression gives t?

Choice	What it suggests	Misconception
A t = d / r	**Correct.**	
B t = dr	Moved r to the other side keeping the operation (multiplication) instead of inverting it.	**INV-OP** Wrong inverse operation
C t = d − r	Undid multiplication by subtracting.	**INV-OP** Wrong inverse operation
D t = r / d	Knows division is needed but divided in the wrong order.	**FRAC-ORDER** Divides in the wrong order

Teaching move. Check with numbers: d = 100 km, r = 50 km/h must give t = 2 h. Which expression gives 2?

NUM · Integers, exponents and rational expressions

NUM01 · Integers, exponents and rational expressions

Difficulty 2 (hypothesized) · Form A #11

Diagnoses: Apply order of operations with a negative sign and an exponent · **CCSS:** 8.EE.A.1, N-RN.A.2 · **NAEP focus:** variables, expressions and operations - order of operations with exponents

What is the value of -3^2?

Choice	What it suggests	Misconception
A 9	Squared −3 as a whole: $(-3)^2$ instead of $-(3^2)$.	**EXP-NEG-BASE** $-a^2$ read as $(-a)^2$
B −9	**Correct.**	
C −6	Read the exponent as multiplication: $-3 \cdot 2$.	**EXP-MULT** Exponent treated as multiplication ($x^3 = 3x$, $2^{-3} = -6$)
D 6	Read the exponent as multiplication and also took the negative as part of the base.	**EXP-MULT** Exponent treated as multiplication ($x^3 = 3x$, $2^{-3} = -6$)

Teaching move. Read -3^2 aloud as 'the opposite of three squared'; contrast -3^2 and $(-3)^2$ as a worked-example pair.

NUM02 · Integers, exponents and rational expressions

Difficulty 2 (hypothesized) · Form A #12

Diagnoses: Substitute a negative value into a polynomial expression · **CCSS:** 6.EE.A.2c, 7.NS.A.2 · **NAEP focus:** variables, expressions and operations - evaluating expressions

If $x = -2$, what is the value of $x^2 - 3x$?

Choice	What it suggests	Misconception
A −2	Computed $-3 \cdot (-2)$ as −6, giving $4 - 6$.	**SIGN-SUB** Errors with subtracting or multiplying negatives
B 2	Substituted without parentheses so x^2 became $-2^2 = -4$, giving $-4 + 6$.	**EXP-NEG-BASE** $-a^2$ read as $(-a)^2$
C −10	Both errors: x^2 as −4 and $-3x$ as −6.	**EXP-NEG-BASE** $-a^2$ read as $(-a)^2$
D 10	**Correct.**	

Teaching move. Always substitute negatives inside parentheses: $(-2)^2 - 3(-2)$. Build the sign rule for multiplication from a pattern.

NUM03 · Integers, exponents and rational expressions

Difficulty 1 (hypothesized) · Form A #13

Diagnoses: Multiply powers with the same base · **CCSS:** 8.EE.A.1, A-SSE.A.2 · **NAEP focus:** variables, expressions and operations - properties of exponents

Which expression is equivalent to $x^3 \cdot x^4$?

Choice	What it suggests	Misconception
A x^7	**Correct.**	
B x^{12}	Multiplied the exponents (power rule) instead of adding them.	**EXP-PRODUCT-MULT** Multiplies exponents when multiplying powers
C $12x^2$	Read x^3 as $3x$ and x^4 as $4x$, then multiplied.	**EXP-MULT** Exponent treated as multiplication ($x^3 = 3x$, $2^{-3} = -6$)
D $2x^7$	Treated the two powers like added terms, so the 'two x's' became a coefficient of 2.	**LIKE-TERMS-OP** Adding like terms treated as multiplication

Teaching move. Expand both as repeated factors and count; derive the product rule and the power rule side by side.

NUM04 · Integers, exponents and rational expressions

Difficulty 2 (hypothesized) · Form B #11

Diagnoses: Evaluate a power with a negative exponent · **CCSS:** 8.EE.A.1, N-RN.A.1 · **NAEP focus:** variables, expressions and operations - properties of exponents

What is the value of 2^{-3}?

Choice	What it suggests	Misconception
A -8	Believes a negative exponent makes the value negative.	**EXP-NEG-EXP** Negative exponent makes the value negative
B -6	Multiplied base and exponent: $2 \cdot (-3)$.	**EXP-MULT** Exponent treated as multiplication ($x^3 = 3x$, $2^{-3} = -6$)
C 1/8	**Correct.**	
D 1/6	Knows a reciprocal is involved but computed $2 \cdot 3$ instead of 2^3.	**EXP-MULT** Exponent treated as multiplication ($x^3 = 3x$, $2^{-3} = -6$)

Teaching move. Pattern table: $2^3 = 8$, $2^2 = 4$, $2^1 = 2$, $2^0 = 1$ ... each step divides by 2; continue to 2^{-3}.

NUM05 · Integers, exponents and rational expressions

Difficulty 3 (hypothesized) · Form B #12

Diagnoses: Add algebraic fractions with unlike denominators · **CCSS:** A-APR.D.6, 7.NS.A.1 · **NAEP focus:** variables, expressions and operations - operations with rational expressions

Which expression is equivalent to x/3 + x/4?

Choice	What it suggests	Misconception
A 2x/7	Added numerators and denominators separately.	**FRAC-ADD-ACROSS** Adds numerators and denominators
B x/7	Added the denominators and kept one numerator.	**FRAC-ADD-ACROSS** Adds numerators and denominators
C 7x/12	**Correct.**	
D $x^2/12$	Used the multiplication rule (multiply across) for addition.	**FRAC-OP-SWAP** Uses the multiplication rule to add fractions

Teaching move. Check at x = 12 (4 + 3 = 7); rewrite with the unit stated: thirds and quarters become twelfths.

LINF · Linear functions and rate of change

LINF01 · Linear functions and rate of change

Difficulty 1 (hypothesized) · Form A #14

Diagnoses: Compute slope from two points · **CCSS:** 8.EE.B.6, F-IF.B.6 · **NAEP focus:** patterns, relations and functions - rate of change

What is the slope of the line through (2, 5) and (6, 13)?

Choice	What it suggests	Misconception
A 2	**Correct.**	
B 1/2	Computed change in x over change in y.	**SLOPE-INVERT** Slope computed as run over rise
C −2	Subtracted the y-values and x-values in opposite orders.	**SLOPE-ORDER** Inconsistent order when subtracting coordinates
D 8	Used only the change in y.	**SLOPE-INCOMPLETE** Slope as change in y only

Teaching move. Define slope as 'change in y when x increases by 1' and read it from a table before using the formula; sketch the points to predict the sign.

LINF02 · Linear functions and rate of change

Difficulty 1 (hypothesized) · Form A #15

Diagnoses: Identify the y-intercept from slope-intercept form · **CCSS:** 8.F.B.4, F-IF.C.7a · **NAEP focus:** algebraic representations - linear equations and graphs

What is the y-intercept of the line $y = 3x - 4$?

Choice	What it suggests	Misconception
A 3	Gave the coefficient of x, without knowing which number plays which role.	**INTERCEPT-COEF** Reads the coefficient of x as the intercept
B −4	**Correct.**	
C 4	Identified the right term but dropped its sign.	**SIGN-DROP** Drops or mishandles a negative sign
D 4/3	Found the x-intercept (where y = 0) instead.	**INTERCEPT-X** Confuses y-intercept with x-intercept

Teaching move. Evaluate at x = 0 and connect the result to where the graph crosses the y-axis.

LINF03 · Linear functions and rate of change

Difficulty 2 (hypothesized) · Form B #15

Diagnoses: Interpret the rate of change of a linear model in context · **CCSS:** F-LE.B.5, 8.F.B.4 · **NAEP focus:** patterns, relations and functions - interpreting linear models

A phone plan costs $C = 15 + 0.10m$ dollars for m minutes of calls. What does 0.10 represent?

Choice	What it suggests	Misconception
A The monthly starting fee	Attached the 'fixed amount' meaning to the rate.	**RATE-START-SWAP** Swaps the rate and the starting amount
B The total cost of m minutes	Confused a per-unit amount with an accumulated amount.	**RATE-TOTAL** Rate confused with total
C The number of minutes used	Confused a fixed parameter with the varying quantity.	**PARAM-VAR** Constant confused with variable
D The cost of each minute, in dollars	**Correct.**	

Teaching move. Write the units of every term (0.10 dollars per minute × m minutes = dollars); compute C(0), C(1), C(2) and ask what changes.

LINF04 · Linear functions and rate of change

Difficulty 2 (hypothesized) · Form B #13

Diagnoses: Write a linear equation from a table · **CCSS:** 8.F.B.4, F-LE.A.2 · **NAEP focus:** algebraic representations - translating between tables and equations

The table shows values of a linear function: $x = 0, 1, 2, 3$ gives $y = 7, 4, 1, -2$. Which equation fits the table?

Choice	What it suggests	Misconception
A $y = 3x + 7$	Found the size of the change but not its direction (y decreases).	**SLOPE-SIGN** Ignores direction of change
B $y = 7x - 3$	Swapped the starting value and the rate of change.	**RATE-START-SWAP** Swaps the rate and the starting amount
C $y = -3x + 7$	**Correct.**	
D $y = -3x$	Assumed the relationship is proportional and ignored the starting value.	**PROPORTIONAL-ALL** Every linear relationship treated as proportional

Teaching move. Before writing any equation, state 'when x = 0, y = __' and 'as x goes up 1, y goes __ by __'.

LINF05 · Linear functions and rate of change Difficulty 2 (hypothesized) · Form B #14

Diagnoses: Interpret a segment of a distance-time graph · **CCSS:** F-IF.B.4, 8.F.B.5 · **NAEP focus:** patterns, relations and functions - interpreting graphs in context

A graph shows Maya's distance from home, d (km), against time, t (hours). From t = 1 to t = 2 the graph is a horizontal segment at d = 3. What was Maya doing during that hour?

Choice	What it suggests	Misconception
A Staying in one place, 3 km from home	**Correct.**	
B Walking along a flat road	Read the shape of the graph as the shape of the road.	**GRAPH-PICTURE** Graph read as a picture of the situation
C Moving at a constant speed of 3 km per hour	Read the distance graph as if it showed speed.	**GRAPH-VARIABLE** Graph read as showing a different quantity
D Travelling 3 km during that hour	Read the height of the graph (position) as distance travelled during the interval.	**RATE-TOTAL** Rate confused with total

Teaching move. Act out distance-time graphs with a walk and narrate each segment; always state what the vertical axis measures.

FUNC · Functions and notation

FUNC01 · Functions and notation

Difficulty 1 (hypothesized) · Form A #16

Diagnoses: Evaluate a function given in function notation · **CCSS:** F-IF.A.2 · **NAEP focus:** patterns, relations and functions - function notation

If $f(x) = x^2 - 1$, what is $f(3)$?

Choice	What it suggests	Misconception
A 5	Read x^2 as $2x$: $2 \cdot 3 - 1$.	**EXP-MULT** Exponent treated as multiplication ($x^3 = 3x$, $2^{-3} = -6$)
B 4	Subtracted before squaring: $(3 - 1)^2$.	**ORDER-OPS** Order of operations not applied to algebraic expressions
C $3x^2 - 3$	Read f(3) as '3 times the rule' instead of 'the output when the input is 3'.	**FN-NOTATION** f(x) read as arithmetic on f
D 8	**Correct.**	

Teaching move. Write $f(\square) = \square^2 - 1$ with an empty box and put the input in every box.

FUNC02 · Functions and notation

Difficulty 3 (hypothesized) · Form A #17

Diagnoses: Evaluate a function at an expression input · **CCSS:** F-IF.A.2, F-BF.B.3 · **NAEP focus:** patterns, relations and functions - function notation

If $f(x) = 2x + 3$, which expression is equal to $f(x + 1)$?

Choice	What it suggests	Misconception
A $2x + 4$	Believes adding 1 to the input adds 1 to the output: $f(x) + 1$.	**FN-SHIFT** $f(x + 1) = f(x) + 1$
B $2x + 5$	**Correct.**	
C $2x^2 + 5x + 3$	Multiplied the rule by $(x + 1)$, reading the parentheses as multiplication.	**FN-NOTATION** f(x) read as arithmetic on f
D $3x + 4$	Appended the input to the rule: $f(x) + x + 1$.	**FN-NOTATION** f(x) read as arithmetic on f

Teaching move. Table of $f(x)$ and $f(x + 1)$ for $x = 0, 1, 2$; compare $f(2 + 1)$ with $f(2) + 1$ numerically.

FUNC03 · Functions and notation

Difficulty 2 (hypothesized) · Form A #18

Diagnoses: Apply the definition of a function · **CCSS:** F-IF.A.1, 8.F.A.1 · **NAEP focus:** patterns, relations and functions - definition of function

In which choice is y a function of x?

Choice	What it suggests	Misconception
A {(1, 4), (2, 4), (3, 4)}	**Correct.**	
B {(4, 1), (4, 2), (4, 3)}	Checked that outputs don't repeat instead of checking inputs; may also reject A because its outputs repeat.	**FN-REVERSE** Input and output reversed
C $x = y^2$	Treats any equation as a function.	**FN-EQUATION** Any equation is a function
D The line x = 3	Believes every straight line is a function.	**FN-LINE** Every straight line is a function

Teaching move. Use the language 'one input, one output'; for each choice, find an input with two outputs, or show that none exists.

FUNC04 · Functions and notation

Difficulty 2 (hypothesized) · Form B #16

Diagnoses: Determine the domain of a rational function · **CCSS:** F-IF.B.5 · **NAEP focus:** patterns, relations and functions - domain and range

What is the domain of $f(x) = 1 / (x - 4)$?

Choice	What it suggests	Misconception
A All real numbers except 0	Remembers 'no dividing by zero' as 'x cannot be zero'.	**DOMAIN-ZERO** Zero is always excluded from the domain
B All real numbers greater than 4	Borrowed the 'must be positive' restriction from square roots.	**DOMAIN-POSITIVE** Domain restrictions borrowed from square roots
C All real numbers except 4	**Correct.**	
D All real numbers except −4	Solved $x - 4 = 0$ with a sign error.	**SIGN-DROP** Drops or mishandles a negative sign

Teaching move. Ask 'which input makes the denominator zero?' and solve it; evaluate f(0) and f(3) to show they are defined.

FUNC05 · Functions and notation Difficulty 1 (hypothesized) · Form B #17

Diagnoses: Interpret function notation as an input-output pair · **CCSS:** F-IF.A.2, F-IF.A.1 · **NAEP focus:** patterns, relations and functions - function notation

For a function h, h(2) = 7. What does this statement mean?

Choice	What it suggests	Misconception
A When the input is 7, the output is 2.	Reversed input and output.	**FN-REVERSE** Input and output reversed
B When the input is 2, the output is 7.	**Correct.**	
C h times 2 equals 7, so h = 3.5.	Read h(2) as multiplication.	**FN-NOTATION** f(x) read as arithmetic on f
D The rule for h is 'add 5'.	Inferred a whole rule from a single input-output pair.	**FN-SINGLEPOINT** Rule inferred from one input-output pair

Teaching move. Find two different rules that both give h(2) = 7; always label pairs (input, output).

MOD · Translating words into symbols

MOD01 · Translating words into symbols

Difficulty 2 (hypothesized) · Form A #19

Diagnoses: Write an equation for a multiplicative comparison · **CCSS:** A-CED.A.1, A-CED.A.2 · **NAEP focus:** algebraic representations - writing equations from verbal descriptions

At a university there are six times as many students as professors. Let S be the number of students and P the number of professors. Which equation is correct?

Choice	What it suggests	Misconception
A 6S = P	Wrote the symbols in word order; S and P are acting as labels, not numbers.	**MOD-REVERSAL** Students-and-professors reversal
B S = P + 6	Read 'six times as many' as 'six more than'.	**MOD-ADDITIVE** Multiplicative comparison read as additive
C S = 6P	**Correct.**	
D S · P = 6	Mapped 'times' to multiplying the two letters.	**MOD-KEYWORD** Keyword translation

Teaching move. Test with numbers first: 10 professors means 60 students. Which equation is true for S = 60, P = 10?

MOD02 · Translating words into symbols

Difficulty 1 (hypothesized) · Form A #20

Diagnoses: Translate a verbal expression with grouping · **CCSS:** 6.EE.A.2a, A-SSE.A.1 · **NAEP focus:** algebraic representations - writing expressions from verbal descriptions

Which expression represents 'twice the sum of a number n and 3'?

Choice	What it suggests	Misconception
A 2n + 3	Translated word by word and missed the grouping implied by 'the sum'.	**MOD-STRUCTURE** Word-order translation misses grouping
B 2 + n + 3	Read 'twice' as 'add 2'.	**MOD-KEYWORD** Keyword translation
C 2 · n · 3	Read 'sum' as multiplication.	**MOD-OP-SWAP** Operation word misread
D 2(n + 3)	**Correct.**	

Teaching move. Ask 'what is the last thing you do?' (double) and build the expression from the outside in; test with n = 5.

MOD03 · Translating words into symbols

Difficulty 2 (hypothesized) · Form B #20

Diagnoses: Represent a percent increase multiplicatively · **CCSS:** A-SSE.A.1b, 7.RP.A.3 · **NAEP focus:** algebraic representations - expressions for percent change

The price p of a jacket increases by 20%. Which expression gives the new price?

Choice	What it suggests	Misconception
A 1.2p	**Correct.**	
B p + 20	Treated 20% as an amount of 20 dollars.	**PCT-ABSOLUTE** Percent treated as an absolute amount
C 0.2p	Computed the increase, not the new price.	**PCT-PART-ONLY** Computes the change, not the new amount
D p / 1.2	Sees the multiplicative structure but applied it in the wrong direction.	**PCT-DIRECTION** Percent change applied in the wrong direction

Teaching move. Double number line with 100% = p; mark 120%. Check with p = $50 and estimate first: the new price must be bigger.

MOD04 · Translating words into symbols

Difficulty 1 (hypothesized) · Form B #18

Diagnoses: Write a linear expression with a fixed amount and a rate · **CCSS:** A-CED.A.1, F-LE.A.2 · **NAEP focus:** algebraic representations - linear models from context

A car rental costs $40 plus $25 per day. Which expression gives the total cost, in dollars, for d days?

Choice	What it suggests	Misconception
A 25 + 40d	Swapped the fixed fee and the daily rate.	**RATE-START-SWAP** Swaps the rate and the starting amount
B 40 + 25d	**Correct.**	
C 65d	Added the two amounts and attached d to the total.	**VAR-CONJOIN** Conjoining unlike terms
D 40d + 25d	Applied 'per day' to both amounts.	**PROPORTIONAL-ALL** Every linear relationship treated as proportional

Teaching move. Ask what it costs for 0, 1 and 2 days before writing anything; label each number with its units.

MOD05 · Translating words into symbols

Difficulty 1 (hypothesized) · Form B #19

Diagnoses: Translate 'less than' correctly · **CCSS:** 6.EE.A.2a, A-SSE.A.1 · **NAEP focus:** algebraic representations - writing expressions from verbal descriptions

Which expression represents 'three less than twice a number n'?

Choice	What it suggests	Misconception
A $3 - 2n$	Wrote the subtraction in word order.	**MOD-ORDER-SUB** Reverses the order in "less than"
B $2(n - 3)$	Placed the grouping on the wrong part of the phrase.	**MOD-STRUCTURE** Word-order translation misses grouping
C $3 < 2n$	Mapped the words 'less than' to the symbol <.	**MOD-KEYWORD** Keyword translation
D $2n - 3$	**Correct.**	

Teaching move. Numeric translation first ('three less than 10 is 7, so 10 − 3'), then replace 10 with 2n.

Appendix A · Misconception catalog

Each code lists what it suggests about a student's thinking, a teaching move, and a quick probe to confirm the diagnosis. Bracketed keys refer to Appendix C.

EQ-OP · Equal sign as "the answer comes next"

Suggests: Reads "=" as a signal to compute, not as a statement that two sides have the same value. Equations become instructions read left to right.

Move: Use true/false and open number sentences with operations on both sides (8 + 4 = n + 5; 47 + 38 = 48 + 37). Ask students to justify without computing. Make "is this statement true?" the routine question before "what is the answer?"

Probe: Is 7 = 3 + 4 true? Is 3 + 4 = 4 + 3 true? Explain.

Options: EQ02B, EQ02D **Research:** [kieran1981], [knuth2006], [carpenter2003]

EQ-ONESIDE · Operation applied to only part of the equation

Suggests: Knows "do something to get x alone" but not that each step must keep both sides equal. Applies the step to the most visible term only.

Move: Pair each solving step with a check by substitution of the original solution into the new equation. If the new equation is false, the step broke equality. Use balance or tape diagrams for multi-term sides.

Probe: x = 5 solves 4x − 8 = 12. Does x = 5 solve x − 8 = 3? What went wrong?

Options: EQ01D, EQ03A, EQ03D, LIN03B **Research:** [kieran1981], [booth2014]

EQ-TRANSPOSE · "Move it to the other side" without the inverse operation

Suggests: Has memorized a move rule ("change sides") without the reason (subtract from both sides). Sign is kept, or changed inconsistently, when a term crosses.

Move: Ban the word "move" for a week. Write the both-sides operation explicitly (−7 on each side) and show the incorrect and correct versions side by side. Ask students to explain why the incorrect one changes the solution.

Probe: Solve 7 − 2x = 15, writing the operation you apply to BOTH sides on each line.

Options: EQ01B, EQ03C, EQ04A, EQ04C, LIN01B, LIN02D, LIN03D, LIN04C **Research:** [booth2014], [kirshner2004]

EQ-UNIQUE · Every equation has exactly one numerical solution

Suggests: Treats solving as producing a single number. Can't interpret 0 = 0 (identity) or 0 = 5 (contradiction), and may test a single value and stop.

Move: Sort equations into one / none / infinitely many solutions by substituting several values. Connect identities to "these two expressions are equivalent" from expression work.

Probe: Try x = 0, 1, 10 in 2(x + 3) = 2x + 6. What do you notice?

Options: EQ05A, EQ05B, EQ05D **Research:** [knuth2006]

EQ-BOTHSIDES · Unknown on both sides cannot be solved

Suggests: Solving is "undo operations on the number until x is alone". When x appears on both sides there is no starting point, so the student declares no solution.

Move: Present x on both sides as a balance. Removing the same x-amount from each side keeps it level. Start with cases where one side's x-term is a whole multiple of the other.

Probe: Is x = 5 a solution of 3(x − 2) = x + 4? Check it.

Options: LIN04D **Research:** [kieran1981]

VAR-CONJOIN · Conjoining unlike terms

Suggests: Expects an algebraic answer to be a single term ("closure"), so 2x + 3 becomes 5x or 7 − 2x becomes 5x.

Move: Substitution check. Evaluate 2x + 3 and 5x at x = 10 and compare. Use algebra tiles or "3 apples + 2 bananas" only as a bridge to the substitution check, not as the explanation; see VAR-LABEL.

Probe: Is 2x + 3 equal to 5x when x = 10? When x = 1? So are they equivalent?

Options: EQ04B, VAR02B, DIST01D, DIST04C, LIN01C, MOD04C **Research:** [booth1988], [kuchemann1981]

VAR-CONCAT · Concatenation (3x with x = 6 read as 36)

Suggests: Brings the place-value meaning of juxtaposed digits into algebra, where juxtaposition means multiplication.

Move: Write the multiplication sign explicitly during substitution (3 · 6), then fade it. Contrast "36" and "3x when x = 6" in a worked example.

Probe: If x = 6, what is 3x? Write it with a multiplication sign.

Options: VAR01C **Research:** [booth1988], [matz1982]

VAR-NOT-NUMBER · Letter treated as not standing for a number

Suggests: Ignores the letter or deletes it ("4x − x = 4"), treating x as a removable symbol rather than a quantity.

Move: Read every expression aloud as a quantity ("four of x, take away one of x"). Use tables of values to show what the expression produces as x changes.

Probe: Complete the table for 4x − x with x = 1, 2, 3. What pattern do you see?

Options: VAR01A, VAR01D, VAR03D **Research:** [kuchemann1981], [booth1988]

VAR-LABEL · Letter as object or label ("e for even", "a for apples")

Suggests: Reads letters as abbreviations for things or properties rather than as numbers that vary.

Move: Insist on precise definitions ("let a = the NUMBER of apples"). Give items where the letter's initial is a trap (s for students vs. the number of students).

Probe: In 'let e be any even number', what kind of thing is e? Name three values it could be.

Options: VAR04B **Research:** [kuchemann1981], [clement1982]

VAR-DIFFERENT · Different letters must have different values

Suggests: Believes distinct letters must name distinct numbers, so a + b + c = a + p + c "can never" be true.

Move: Have students find values that make the statement true, and values that make it false. Introduce "always / sometimes / never" as a routine with a required example for "sometimes".

Probe: Can b and p both equal 4? Try it in a + b + c = a + p + c.

Options: VAR03B **Research:** [kuchemann1981]

VAR-SURFACE · Judging by surface features of symbols

Suggests: Decides from appearance ("it has a 2 so it is even", "multiplying makes bigger", "same letters on both sides so always true") without reasoning about values.

Move: Counterexample hunts. For each claim, students test negative, zero, fractional and large values. Celebrate a single counterexample as a proof that "always" is false.

Probe: Is 2n always bigger than n + 2? Try n = 1 and n = −3.

Options: EQ02A, VAR03A, VAR04C, VAR05A **Research:** [kuchemann1981], [booth2014]

VAR-SPECIFIC · Generalizing from a single example

Suggests: Tests one convenient value (often n = 2) and treats the result as a general truth. Doesn't yet see a variable as standing for all values at once.

Move: Require at least three test values chosen to break the claim (0, 1, a negative, a fraction). Then ask for a reason that holds for every value, not just the tested ones.

Probe: n + 2 was even when n = 2. Is it always even? Choose a value that tests it fairly.

Options: VAR04A, VAR05B, VAR05C **Research:** [kuchemann1981]

VAR-COEF-ADD · Coefficient read as addition (3x as 3 + x)

Suggests: Treats the coefficient as a separate number added to x, so "undoing" 3x means subtracting 3.

Move: Rewrite 3x as x + x + x and as 3 · x. Ask what single operation turns 3x back into x. Contrast with 3 + x.

Probe: If x = 5, what is 3x? What is 3 + x? Are they the same?

Options: EQ01C **Research:** [booth1988]

LIKE-TERMS-OP · Adding like terms treated as multiplication

Suggests: Confuses the rules for combining terms with the rules for multiplying them (3x + 5x → 15x or $8x^2$, x + x → x^2).

Move: Contrast pairs: 3x + 5x vs. 3x · 5x, x + x vs. x · x, each checked by substituting x = 2. Area model for products, length model for sums.

Probe: Evaluate 3x + 5x and $8x^2$ at x = 2.

Options: VAR02C, VAR02D, NUM03D **Research:** [booth2014], [matz1982]

DIST-FIRST · Distributes to the first term only

Suggests: Multiplies the term next to the coefficient and leaves the rest of the parentheses alone.

Move: Area model. A rectangle of width 4 and length x + 3 has two parts, so it has two products. Then use arrows from the factor to every term inside the parentheses, and fade them.

Probe: Draw a rectangle for 4(x + 3). How many pieces does it have?

Options: DIST01A, DIST04B, LIN04A **Research:** [matz1982], [booth2014]

DIST-ADD · Adds the outside factor instead of multiplying

Suggests: Treats the factor outside the parentheses as joining by addition (4(x + 3) → 4x + 7).

Move: Read 4(x + 3) as "four groups of (x + 3)" and write the groups out once: (x + 3) + (x + 3) + (x + 3) + (x + 3).

Probe: Write 4(x + 3) as a sum of four groups. Now simplify.

Options: DIST01B **Research:** [booth1988]

DIST-NEG · Negative not distributed to every term

Suggests: Drops parentheses after a minus sign without changing every sign inside. Reads −(x − 3) as −x − 3.

Move: Rewrite the minus as −1 · ( ), then distribute. Check by substitution (x = 0 is a quick test: 10 − (0 − 3) = 13).

Probe: Evaluate 10 − (x − 3) at x = 0. Does your simplified form give the same?

Options: DIST02B, DIST02C **Research:** [vlassis2004], [booth2014]

DIST-SQUARE · $(a + b)^2 = a^2 + b^2$ (linearity overgeneralization)

Suggests: Extends the distributive pattern to exponents, treating squaring as if it distributes over addition.

Move: Area model of a square with side x + 5 shows four regions, not two. Numeric check: $(3 + 5)^2 = 64$, but $3^2 + 5^2 = 34$.

Probe: Is $(3 + 5)^2$ equal to $3^2 + 5^2$? Draw the square.

Options: DIST03A **Research:** [matz1982]

DIST-PARTIAL · Incomplete expansion (misses a cross term)

Suggests: Knows binomial products have more than two terms but tracks the partial products inconsistently.

Move: Use a 2×2 area grid (box method) for every binomial product until students record all four partial products reliably.

Probe: Fill in a 2×2 grid for $(x + 5)(x + 5)$.

Options: DIST03C **Research:** [matz1982]

DIST-CANCEL · Cancelling across addition in a fraction

Suggests: Treats a common number in numerator and denominator as cancellable even when it is a term, not a factor.

Move: Factor first, then cancel: $(3x + 6)/3 = 3(x + 2)/3$. Numeric check: $x = 1$ gives $9/3 = 3$, not 7.

Probe: Evaluate $(3x + 6)/3$ at $x = 1$. Does $x + 6$ give the same?

Options: DIST04A **Research:** [kirshner2004]

FACT-SQUARE · Difference of squares treated as a perfect square

Suggests: Matches the shape "x^2 … 9" to $(x - 3)^2$ without checking the middle term.

Move: Multiply back as a routine. Every factorization is checked by expanding. Contrast $x^2 - 9$ with $x^2 - 6x + 9$.

Probe: Expand $(x - 3)^2$. Is it $x^2 - 9$?

Options: DIST05A **Research:** [kirshner2004]

FACT-SUM-PRODUCT · Sum/product search misapplied

Suggests: Searches for factor pairs of the constant without matching the middle coefficient (here, zero).

Move: Make the coefficient of x explicit ($x^2 + 0x - 9$) before searching for the pair.

Probe: Rewrite $x^2 - 9$ with an x-term. What must the two numbers add to?

Options: DIST05C **Research:** [kirshner2004]

FACT-COMMON · Pulls out a factor that is not common to every term

Suggests: Factors x out of $x^2 - 9$ as if 9 contained x.

Move: Before factoring out, list each term's factors in a row and highlight only what is shared by all rows.

Probe: Expand $x(x - 9)$. Is it $x^2 - 9$?

Options: DIST05D **Research:** [kirshner2004]

SIGN-DROP · Drops or mishandles a negative sign

Suggests: Loses the sign of a coefficient or result, especially when dividing ($-2x = 8 \rightarrow x = 4$) or when solving $x - 4 = 0$.

Move: Circle each term together with the sign in front of it. Check every answer by substitution.

Probe: Substitute your answer into the original equation. Is it true?

Options: LIN01A, LIN02C, LINF02C, FUNC04D **Research:** [vlassis2004]

SIGN-SUB · Errors with subtracting or multiplying negatives

Suggests: Treats $-3 \cdot (-2)$ as -6, or subtracting a negative as subtracting.

Move: Use a number line with direction for subtraction. Build the rule from patterns ($3 \cdot -2$, $2 \cdot -2$, $1 \cdot -2$, $0 \cdot -2$, $-1 \cdot -2$ …) rather than stating it.

Probe: Continue the pattern: $2 \cdot (-2) = -4$, $1 \cdot (-2) = -2$, $0 \cdot (-2) = 0$, $(-1) \cdot (-2) = ?$

Options: DIST02D, NUM02A **Research:** [vlassis2004]

INV-OP · Wrong inverse operation

Suggests: Undoes an operation with the wrong inverse (undoes ÷3 by ÷3, or undoes ×r by −r).

Move: Arrow diagrams ("x → ×3 → +2 → 17") read forward, then undone backward with inverses written on the arrows.

Probe: Draw the arrow diagram for x/3 + 2 and undo it.

Options: LIN03A, LIN05B, LIN05C **Research:** [kieran1981]

INEQ-FLIP · Inequality direction not reversed after multiplying or dividing by a negative

Suggests: Solves inequalities exactly like equations. The ordering meaning of < is not connected to the steps.

Move: Test a value in the original inequality and in the "answer" (x = −5 makes −3x + 4 > 10 true; does x > −2 include −5?). Derive the flip from the number line, not as a rule.

Probe: Is x = −5 a solution of −3x + 4 > 10? Is it in your answer set?

Options: LIN02B **Research:** [booth2014]

FRAC-ORDER · Divides in the wrong order

Suggests: Knows division is needed but does not track which quantity is the dividend.

Move: Check with numbers. If d = 100 and r = 50, what must t be? Which expression gives 2?

Probe: d = 100, r = 50. Compute d/r and r/d. Which makes sense as a time?

Options: LIN05D **Research:** [kieran1981]

EXP-NEG-BASE · $-a^2$ read as $(-a)^2$

Suggests: Applies the exponent to the negative sign. Order of operations with the unary minus is not secure.

Move: Read -3^2 aloud as "the opposite of three squared". Contrast -3^2 and $(-3)^2$ in a worked-example pair. Substitute negatives in parentheses, always.

Probe: Evaluate -3^2 and $(-3)^2$. Explain the difference in words.

Options: NUM01A, NUM02B, NUM02C **Research:** [cangelosi2013]

EXP-MULT · Exponent treated as multiplication ($x^3 = 3x$, $2^{-3} = -6$)

Suggests: Reads the exponent as a factor rather than a count of repeated factors.

Move: Expand into repeated factors ($x^3 = x \cdot x \cdot x$) every time until fluent. Contrast 2^3 and $2 \cdot 3$.

Probe: Write 2^3 and $3 \cdot 2$ as repeated operations. Are they equal?

Options: DIST03D, NUM01C, NUM01D, NUM03C, NUM04B, NUM04D, FUNC01A **Research:** [cangelosi2013]

EXP-PRODUCT-MULT · Multiplies exponents when multiplying powers

Suggests: Confuses the product rule (add exponents) with the power rule (multiply exponents).

Move: Expand both: $x^3 \cdot x^4$ is seven x's in a row, while $(x^3)^4$ is four groups of three. Derive both rules from expansion side by side.

Probe: Write $x^3 \cdot x^4$ as repeated factors. Count the x's.

Options: NUM03B **Research:** [matz1982]

EXP-NEG-EXP · Negative exponent makes the value negative

Suggests: Associates the negative sign in the exponent with a negative result.

Move: Pattern table: $2^3 = 8$, $2^2 = 4$, $2^1 = 2$, $2^0 = 1$, $2^{-1} = ½$ … each step divides by 2.

Probe: Continue the pattern 8, 4, 2, 1, … for 2^{-1}, 2^{-2}, 2^{-3}.

Options: NUM04A **Research:** [cangelosi2013]

ORDER-OPS · Order of operations not applied to algebraic expressions

Suggests: Applies operations in the order they are written, or subtracts before squaring.

Move: Rewrite each expression with explicit grouping before substitution. Ask "what is the last operation?" to identify the structure.

Probe: In $x^2 - 1$, what is the last operation performed?

Options: FUNC01B **Research:** [kirshner2004]

FRAC-ADD-ACROSS · Adds numerators and denominators

Suggests: Adds fractions componentwise, so the meaning of a denominator as a unit size is not active.

Move: Numeric check (1/2 + 1/2 is not 2/4). Rewrite to a common denominator with the unit stated ("thirds and quarters → twelfths").

Probe: Is 1/2 + 1/2 equal to 2/4?

Options: NUM05A, NUM05B **Research:** [booth2014]

FRAC-OP-SWAP · Uses the multiplication rule to add fractions

Suggests: Applies the "multiply across" procedure regardless of the operation.

Move: Contrast pairs of addition and multiplication with the same fractions, checked by substituting x = 12.

Probe: With x = 12, evaluate x/3 + x/4 and (x/3)(x/4).

Options: NUM05D **Research:** [booth2014]

SLOPE-INVERT · Slope computed as run over rise

Suggests: Knows slope is a ratio of differences but not which difference is per which. The meaning "change in y per 1 unit of x" is not active.

Move: Define slope as a rate ("how much y changes when x increases by 1") and read it off a table before any formula.

Probe: When x goes up by 1, how much does y go up between (2, 5) and (6, 13)?

Options: LINF01B **Research:** [teuscher2010], [leinhardt1990]

SLOPE-ORDER · Inconsistent order when subtracting coordinates

Suggests: Subtracts y-values in one order and x-values in the other, producing the wrong sign.

Move: Label points 1 and 2 and keep the same order in the numerator and denominator. Sketch the points first to predict the sign.

Probe: Before computing, does the line go up or down from left to right?

Options: LINF01C **Research:** [teuscher2010]

SLOPE-INCOMPLETE · Slope as change in y only

Suggests: Equates slope with how much y changes, ignoring the change in x.

Move: Compare two lines with the same rise and different runs. Ask which is steeper and why.

Probe: Two lines both rise 8. One runs 4, one runs 8. Same slope?

Options: LINF01D **Research:** [teuscher2010]

SLOPE-SIGN · Ignores direction of change

Suggests: Finds the size of the rate but not whether y increases or decreases.

Move: Before writing an equation from a table, students state "as x goes up, y goes ___" and check that the sign of the slope matches.

Probe: As x goes up in the table, does y go up or down?

Options: LINF04A **Research:** [leinhardt1990]

INTERCEPT-COEF · Reads the coefficient of x as the intercept

Suggests: Identifies "the number in the equation" without knowing which parameter plays which role.

Move: Evaluate the function at x = 0 and connect the result to the graph's crossing point.

Probe: What is y when x = 0 in y = 3x − 4?

Options: LINF02A **Research:** [leinhardt1990]

INTERCEPT-X · Confuses y-intercept with x-intercept

Suggests: Knows "intercept" means a crossing point but not which axis is meant.

Move: Graph and table together. The y-intercept is the row with x = 0, and the x-intercept is the row with y = 0.

Probe: Which row of a table shows the y-intercept?

Options: LINF02D **Research:** [leinhardt1990]

RATE-START-SWAP · Swaps the rate and the starting amount

Suggests: Recognizes a linear model has two parameters but attaches the meanings to the wrong numbers.

Move: In context, ask "what does it cost for 0 days / minutes?" and "what changes each additional day?" before writing anything. Annotate each number in the model with units (dollars, dollars per day).

Probe: How much would the rental cost for 0 days? For 1 day? For 2?

Options: LINF03A, LINF04B, MOD04A **Research:** [leinhardt1990]

RATE-TOTAL · Rate confused with total

Suggests: Does not distinguish a per-unit amount from an accumulated amount.

Move: Units analysis. Write the units of every term (0.10 dollars per minute × m minutes = dollars).

Probe: What are the units of 0.10 in C = 15 + 0.10m?

Options: LINF03B, LINF05D **Research:** [teuscher2010]

PARAM-VAR · Constant confused with variable

Suggests: Cannot separate the fixed numbers in a model from the quantity that varies.

Move: Vary the input in a table and ask which numbers in the formula changed and which stayed the same.

Probe: If m changes from 100 to 200, which numbers in C = 15 + 0.10m change?

Options: LINF03C **Research:** [kuchemann1981]

PROPORTIONAL-ALL · Every linear relationship treated as proportional

Suggests: Assumes y = mx and ignores the starting value.

Move: Check the x = 0 row of every table first. Contrast proportional and non-proportional tables side by side.

Probe: What is y when x = 0 in the table? What does y = −3x give at x = 0?

Options: LINF04D, MOD04D **Research:** [leinhardt1990]

GRAPH-PICTURE · Graph read as a picture of the situation

Suggests: Interprets the shape of the graph as the shape of the physical path (flat line = flat road).

Move: Have students act out graphs (walk to match a distance-time graph) and narrate. Always state "the vertical axis shows ___".

Probe: What does the vertical axis measure? What happens to that quantity from t = 1 to t = 2?

Options: LINF05B **Research:** [bell1981], [leinhardt1990]

GRAPH-VARIABLE · Graph read as showing a different quantity

Suggests: Reads a distance-time graph as if it showed speed, so a flat segment means constant speed.

Move: Pair distance-time and speed-time graphs of the same trip, and match segments between them.

Probe: If the distance does not change for an hour, what is the speed?

Options: LINF05C **Research:** [bell1981], [leinhardt1990]

FN-NOTATION · f(x) read as arithmetic on f

Suggests: Treats f(3) as f × 3 or f(x + 1) as an operation on the rule, rather than as "the output when the input is 3".

Move: Use an input-output machine. Write f(□) = □² − 1 with a blank box, and substitute whatever goes in the box, including expressions.

Probe: f(x) = x² − 1. Fill in: f(□) = □² − 1. Now put 3 in the box.

Options: FUNC01C, FUNC02C, FUNC02D, FUNC05C **Research:** [oehrtman2008]

FN-SHIFT · f(x + 1) = f(x) + 1

Suggests: Thinks changing the input by 1 changes the output by 1. Input and output transformations are confused.

Move: Table of f(x) and f(x + 1) side by side for several x. Contrast f(x + 1) with f(x) + 1 numerically.

Probe: f(x) = 2x + 3. Compute f(2 + 1) and f(2) + 1.

Options: FUNC02A **Research:** [oehrtman2008]

FN-REVERSE · Input and output reversed

Suggests: Reads h(2) = 7 as input 7, output 2. In definitions, checks repeated outputs instead of repeated inputs.

Move: Always label pairs as (input, output) and arrows input → output. Use "one input, one output" machine language.

Probe: In h(2) = 7, which number went into the machine?

Options: FUNC03B, FUNC05A **Research:** [oehrtman2008]

FN-EQUATION · Any equation is a function

Suggests: Equates "function" with "has a formula", rather than with each input having exactly one output.

Move: For x = y², ask "if x = 4, what is y?" and find two answers. Graph it and apply the definition directly.

Probe: In x = y², how many y-values go with x = 4?

Options: FUNC03C **Research:** [oehrtman2008]

FN-LINE · Every straight line is a function

Suggests: Has generalized "linear functions" to "all lines", so the vertical line x = 3 is accepted.

Move: List points on x = 3 and check one-input-one-output directly.

Probe: Name three points on the line x = 3. What do their x-values have in common?

Options: FUNC03D **Research:** [oehrtman2008]

FN-SINGLEPOINT · Rule inferred from one input-output pair

Suggests: Treats one data point as determining the whole function.

Move: Generate several different rules through the same point (2, 7). Discuss what extra information pins a function down.

Probe: Find two different rules with h(2) = 7.

Options: FUNC05D **Research:** [oehrtman2008]

DOMAIN-ZERO · Zero is always excluded from the domain

Suggests: Remembers "you can't divide by zero" as "x can't be zero", rather than "the denominator can't be zero".

Move: Ask "what input makes the denominator 0?" and solve it explicitly.

Probe: What is f(0) for f(x) = 1/(x − 4)? Is it defined?

Options: FUNC04A **Research:** [oehrtman2008]

DOMAIN-POSITIVE · Domain restrictions borrowed from square roots

Suggests: Applies a "must be positive" restriction from square-root functions to rational functions.

Move: Evaluate the function at inputs below the suspected boundary (f(0) = −1/4) to show they are defined.

Probe: Is f(0) defined for f(x) = 1/(x − 4)?

Options: FUNC04B **Research:** [oehrtman2008]

MOD-REVERSAL · Students-and-professors reversal

Suggests: Writes symbols in word order ("six times students… professors" → 6S = P), treating letters as labels rather than numbers.

Move: Test with numbers. If there are 10 professors, how many students? Does the equation give that? Write the equation only after the numeric check.

Probe: If P = 10, what should S be? Does 6S = P give that?

Options: MOD01A **Research:** [clement1982]

MOD-ADDITIVE · Multiplicative comparison read as additive

Suggests: Interprets "six times as many" as "six more than".

Move: Contrast "6 more than" and "6 times as many" with bar models at two different sizes.

Probe: With 10 professors, how many students is '6 more than' and how many is '6 times as many'?

Options: MOD01B **Research:** [clement1982]

MOD-KEYWORD · Keyword translation

Suggests: Maps key words to operations or symbols ("times" → multiply the letters, "less than" → <, "twice" → +2) without modeling the relationship.

Move: Retire keyword charts. Have students describe the quantities and the relationship first, then test the expression with a number.

Probe: Pick a number for n. What is '3 less than twice n'? Does your expression give that?

Options: MOD01D, MOD02B, MOD05C **Research:** [clement1982]

MOD-STRUCTURE · Word-order translation misses grouping

Suggests: Translates phrase by phrase, so "twice the sum of n and 3" becomes 2n + 3, or the grouping is placed on the wrong part.

Move: Ask for the last operation in the phrase ("twice the sum" → the last operation is doubling) and build the expression from the outside in.

Probe: In 'twice the sum of n and 3', what is the last thing you do?

Options: MOD02A, MOD05B **Research:** [clement1982]

MOD-OP-SWAP · Operation word misread

Suggests: Interprets "sum" or "plus" as multiplication, or the reverse.

Move: Quick numeric translation drills ("the sum of 4 and 3 is ___") before introducing variables.

Probe: What is the sum of 4 and 3? The product?

Options: MOD02C **Research:** [clement1982]

MOD-ORDER-SUB · Reverses the order in "less than"

Suggests: Writes "3 less than 2n" as 3 − 2n, following word order.

Move: Numeric translation first ("3 less than 10 is 7, so 10 − 3"), then replace 10 by the expression.

Probe: What is 3 less than 10? Write the subtraction.

Options: MOD05A **Research:** [clement1982]

PCT-ABSOLUTE · Percent treated as an absolute amount

Suggests: Adds 20 (or 0.20) to the price rather than 20% of the price.

Move: Use a double number line with 100% = p and mark 120%. Check with p = $50.

Probe: If p = $50, what is the price after a 20% increase?

Options: MOD03B **Research:** [parker1995]

PCT-PART-ONLY · Computes the change, not the new amount

Suggests: Finds 20% of p correctly but reports the increase instead of the new total.

Move: Ask "what does this expression give you, the increase or the new price?" and annotate both.

Probe: With p = $50, what does 0.2p equal? Is that the new price?

Options: MOD03C **Research:** [parker1995]

PCT-DIRECTION · Percent change applied in the wrong direction

Suggests: Divides by 1.2 (undoing an increase) when it should multiply. Multiplicative structure is partly in place but the direction is lost.

Move: Estimate first. After an increase the new price must be larger. Check whether the expression gives a larger value.

Probe: Is p/1.2 bigger or smaller than p?

Options: MOD03D **Research:** [parker1995]

Appendix B · Answer keys and scoring grids

Form A

#	Item	Key	A	B	C	D
1	EQ01	A	✓	EQ-TRANSPOSE	VAR-COEF-ADD	EQ-ONESIDE
2	EQ02	C	VAR-SURFACE	EQ-OP	✓	EQ-OP
3	EQ03	B	EQ-ONESIDE	✓	EQ-TRANSPOSE	EQ-ONESIDE
4	VAR01	B	VAR-NOT-NUMBER	✓	VAR-CONCAT	VAR-NOT-NUMBER
5	VAR02	A	✓	VAR-CONJOIN	LIKE-TERMS-OP	LIKE-TERMS-OP
6	DIST01	C	DIST-FIRST	DIST-ADD	✓	VAR-CONJOIN
7	DIST02	A	✓	DIST-NEG	DIST-NEG	SIGN-SUB
8	DIST03	B	DIST-SQUARE	✓	DIST-PARTIAL	EXP-MULT
9	LIN01	D	SIGN-DROP	EQ-TRANSPOSE	VAR-CONJOIN	✓
10	LIN02	A	✓	INEQ-FLIP	SIGN-DROP	EQ-TRANSPOSE
11	NUM01	B	EXP-NEG-BASE	✓	EXP-MULT	EXP-MULT
12	NUM02	D	SIGN-SUB	EXP-NEG-BASE	EXP-NEG-BASE	✓
13	NUM03	A	✓	EXP-PRODUCT-MULT	EXP-MULT	LIKE-TERMS-OP
14	LINF01	A	✓	SLOPE-INVERT	SLOPE-ORDER	SLOPE-INCOMPLETE
15	LINF02	B	INTERCEPT-COEF	✓	SIGN-DROP	INTERCEPT-X
16	FUNC01	D	EXP-MULT	ORDER-OPS	FN-NOTATION	✓
17	FUNC02	B	FN-SHIFT	✓	FN-NOTATION	FN-NOTATION
18	FUNC03	A	✓	FN-REVERSE	FN-EQUATION	FN-LINE
19	MOD01	C	MOD-REVERSAL	MOD-ADDITIVE	✓	MOD-KEYWORD
20	MOD02	D	MOD-STRUCTURE	MOD-KEYWORD	MOD-OP-SWAP	✓

Form B

#	Item	Key	A	B	C	D
1	EQ04	D	EQ-TRANSPOSE	VAR-CONJOIN	EQ-TRANSPOSE	✓
2	EQ05	C	EQ-UNIQUE	EQ-UNIQUE	✓	EQ-UNIQUE
3	VAR04	D	VAR-SPECIFIC	VAR-LABEL	VAR-SURFACE	✓
4	VAR05	D	VAR-SURFACE	VAR-SPECIFIC	VAR-SPECIFIC	✓
5	VAR03	C	VAR-SURFACE	VAR-DIFFERENT	✓	VAR-NOT-NUMBER
6	DIST04	D	DIST-CANCEL	DIST-FIRST	VAR-CONJOIN	✓
7	DIST05	B	FACT-SQUARE	✓	FACT-SUM-PRODUCT	FACT-COMMON
8	LIN04	B	DIST-FIRST	✓	EQ-TRANSPOSE	EQ-BOTHSIDES
9	LIN05	A	✓	INV-OP	INV-OP	FRAC-ORDER
10	LIN03	C	INV-OP	EQ-ONESIDE	✓	EQ-TRANSPOSE
11	NUM04	C	EXP-NEG-EXP	EXP-MULT	✓	EXP-MULT
12	NUM05	C	FRAC-ADD-ACROSS	FRAC-ADD-ACROSS	✓	FRAC-OP-SWAP
13	LINF04	C	SLOPE-SIGN	RATE-START-SWAP	✓	PROPORTIONAL-ALL
14	LINF05	A	✓	GRAPH-PICTURE	GRAPH-VARIABLE	RATE-TOTAL
15	LINF03	D	RATE-START-SWAP	RATE-TOTAL	PARAM-VAR	✓
16	FUNC04	C	DOMAIN-ZERO	DOMAIN-POSITIVE	✓	SIGN-DROP
17	FUNC05	B	FN-REVERSE	✓	FN-NOTATION	FN-SINGLEPOINT

#	Item	Key	A	B	C	D
18	MOD04	B	RATE-START-SWAP	✓	VAR-CONJOIN	PROPORTIONAL-ALL
19	MOD05	D	MOD-ORDER-SUB	MOD-STRUCTURE	MOD-KEYWORD	✓
20	MOD03	A	✓	PCT-ABSOLUTE	PCT-PART-ONLY	PCT-DIRECTION

Appendix C · References

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